

Problems in Standard Deviation

- 1) Goals scored by two teams A and B in a football season were as follows :

No. of goals scored in a match	No. of matches	
	A	B
0	27	17
1	9	9
2	8	6
3	5	5
4	4	3

Find out which team is more consistent.

Ans:

the coefficient of variation is less for the team B, hence team B is more consistent.

- 2)

The first of the two samples has 100 items with mean 15 and standard deviation

3. If the whole group has 250 items with mean 15.6 and standard deviation $\sqrt{13.44}$, find the standard deviation of the second group.

Ans: 4

- 3)

Compute the standard deviation for the following data relating to marks obtained by 15 students :

12, 21, 21, 23, 27, 28, 30, 34, 37, 39, 39, 39, 40, 49, 54.

- 4)

Calculate mean and standard deviation for the following distribution :

$x :$	56	63	70	77	84	91	98
$f :$	3	6	14	16	13	6	2

- 5)

The mean of five items of an observation is 4 and the variance is 5.2. If three of the items are 1, 2 and 6, then find the other two.

- 6)

Show that the variance of the first n positive integers is $\frac{1}{12} (n^2 - 1)$.

- 7)

A student obtained the mean and standard deviation of 100 observations as 40 and 5 respectively. It was later discovered that he had wrongly copied down an observation as 50 instead of 40. Calculate the correct mean and standard deviation.

- 8)

The following are scores of two batsmen A and B in a series of innings :

A :	12	115	6	73	7	19	119	36	84	29
B :	47	12	16	42	4	51	37	48	43	0

Who is the better run getter and who is more consistent ?

- 9)

The heights and weights of 10 army men are given below. In which characteristic are they more variable ?

Height in cm :	170	172	168	177	179	171	173	178	173	179
Weight in kg :	75	74	75	76	77	73	76	75	74	75

Answers:

- 3) 10.9 4) 76.53, 9.87 5) 4,7 7) 39.9, 4.9 8) A, B 9) Height.